

TABLE II
Sensors, Chips and Other Components Compared

Product	Bluetooth chip	Accelerometer	Other sensors	Memory	Others
Fitbit Flex	• nRF8001 • Nordic Semiconductor • Bluetooth Smart connectivity IC • Integrates Bluetooth Smart v4.0 Radio, link layer, host stack, a simple serial interface, DC/DC regulator and 32kHz RC oscillator	• LIS2DH • STMicroelectronics • MEMS digital-output motion sensor • 3-axis, user-selectable full scales of $\pm 2g/\pm 4g/\pm 8g/\pm 16g$, digital I2C/SPI standard output		• Not specified	• Vibration motor; coin type • BQ24040, charger IC - Texas Instruments
Garmin Vivofit	• Not specified	• ADXL362 • Analog Devices MEMS digital output motion sensor • 3-axis, user-selectable full scales of $\pm 2g/\pm 4g/\pm 8g$, digital SPI standard output • Integrates multi-mode output FIFO, built-in temperature sensor and several activity detection modes including adjustable threshold sleep and wake-up operation		• Not specified	• TPS3420, 2-channel push-button controller with configurable delay - Texas Instruments • M41T62, low-power serial RTC with alarm - STMicroelectronics • LTC3388-1, 20V step-down regulator, Linear Technology • CP2400, LCD controller - Silicon Labs
Microsoft Band	• AR3002 • Qualcomm • All-CMOS, low-power, cost-effective • Bluetooth 4.0 + HS HCI ROM solution	• 3-axis accelerometer	• Optical heart rate sensor, gyrometer, GPS module, ambient light sensor, UV sensor, skin temperature sensor, capacitive sensor and galvanic skin response	• MX29GL640EHXFS-70G • Macronix • 64MB non-volatile NOR parallel flash memory • CY62167EV18 • Cypress Semiconductor • 16Mbit (1M$\times$16-bit) SRAM	• Haptic vibration motor • Microphone
Samsung Gear Fit	• BCM4334 • Single-chip dual-band combo • Bluetooth 4.0 + HS and FM receiver	• MPU-6500 • 6-axis motion-tracking device combining 3-axis accelerometer • 3-axis gyroscope and digital motion processor	• Heart rate sensor	• MX69V28F64 • Macronix read-while-write, multiplexed, burst mode, flash memory • 128Mbit (8M$\times$16-bit) CMOS 1.8V	• MAX77836, MUIC-USB interface circuit plus Li+ battery charger - Maxim Integrated • 8FM006A, touchscreen controller - Melfas • TUSB1211, standalone USB transceiver chip - Texas Instruments

one thing that could make or break a device, irrespective of whether the software and circuits are perfect.

"People will not wear something that is clumsy or looks crude. They will wear and market the product (by word of mouth) only if it looks stylish and futuristic. Hence, a lot of emphasis should be given to the design aspect of the wearable. Style and comfort are all some predominant features that should be focused on. The product should then be built around the design of the final product, and not the other way around," explains Sanjeev.

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Kashinath adds to this, from his own experience, "Since we work on wearables in clothing form factor, our main challenge has been to try to

reduce the presence of 'hard' ware and replicate similar functionality using soft materials like conducting thread and fabric." ●